

END TERM EXAMINATION

FOURTH SEMESTER [B.TECH] JUNE 2024

Paper Code: EEC-206

Subject: Network Analysis & Synthesis

Time: 3 Hours

Maximum Marks: 75

Note: Attempt five questions in all including Q.No.1 which is compulsory. Select one question from each unit. Assume missing data, if any.

Q1 Attempt all questions:

(3x5=15)

- Define graph, tree and co-tree.
- Find laplace transform of function $f(t)=2u(t-1)-tu(t-4)$.
- Define hybrid parameters and draw its equivalent circuit.
- Test whether the polynomial $P(s)$ is Hurwitz or not $P(s)=s^4+7s^3+6s^2+21s+8$?
- Define characteristic impedance, attenuation and cut off frequency for passive filters.

UNIT-I

- Q2 a) Using mesh current analysis, find the drop in the capacitor for the network given in the figure 1 (8)

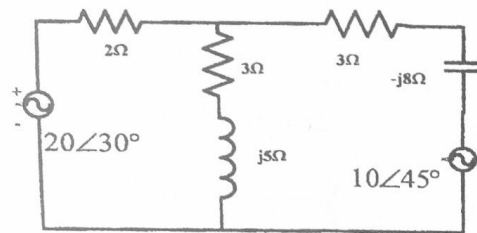


Figure 1

- b) Find the Thevenin's equivalent circuit of the network shown (7)

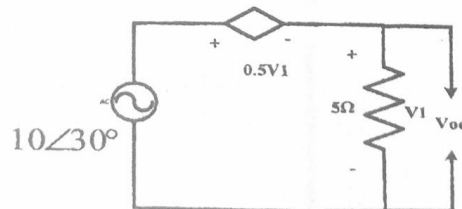


Figure 2

- Q3 Obtain the tree from the graph given in figure 3 with links (5,6,7). Find the incidence matrix, fundamental loop matrix, and fundamental cut-set matrix. (15)

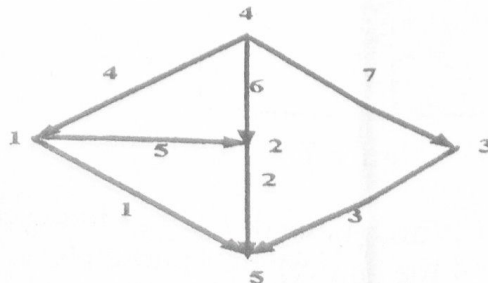


Figure 3

UNIT-II

- Q4 a) The switch in circuit of the figure 4 has been closed for a long time. It opens at $t=0$. Find $V_c(t)$ for $t>0$ using classical method. (9)

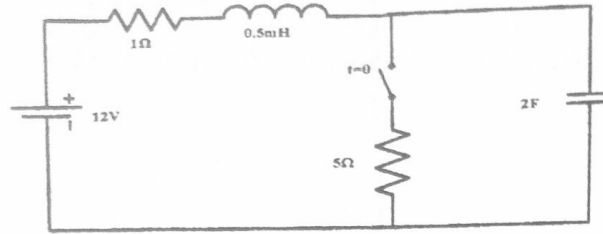


Figure 4

- b) Obtain the $F(t)$ of the periodic waveform shown in figure 5. (6)

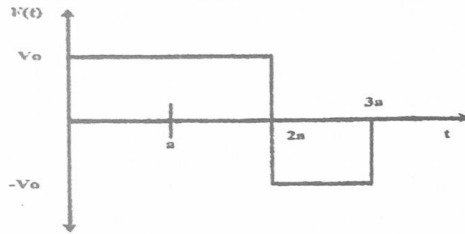


Figure 5

- Q5 a) In the circuit shown in figure 6, the switch S is closed and a steady state is reached. At $t=0$, the switch is opened. Find an expression for $I_L(s)$ current through inductor. (9)

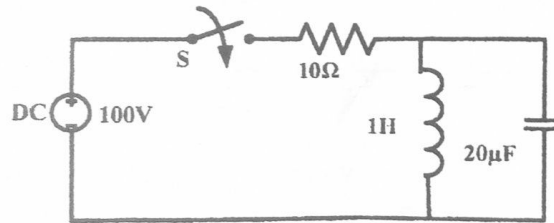


Figure 6

- b) Find the current $i(t) = \frac{5}{s(s^2+s+2)}$. Find $I(s)$ and hence determine the value of $i(0^+)$ and $i(\infty)$. (6)

UNIT-III

- Q6 a) Obtain the Z parameters for the network in Fig. 7. (8)

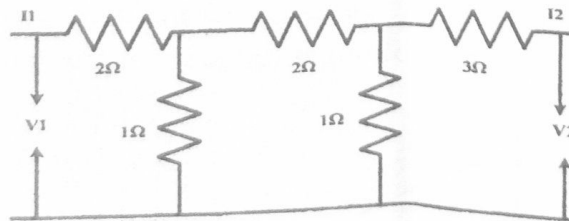


Figure 7

- b) If two 2-port networks are chained together in cascade then derive its condition and find the equivalent T parameters. (7)

- Q7 a) Determine transmission parameters for the network shown in figure 8 using the concept of interconnection of four two port networks N_1 , N_2 , N_3 , and N_4 in cascade. (8)

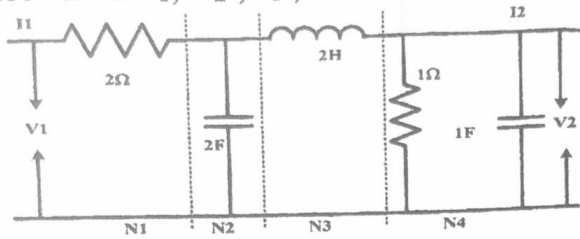


Figure 8

- b) For the π network of the figure 9, obtain the Y-parameters (where Y_a , Y_b and Y_c are admittances). (7)

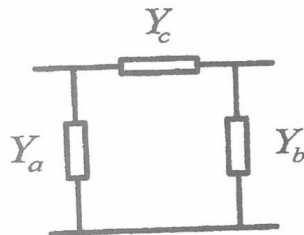


Figure 9

UNIT-IV

- Q8 a) Synthesize the network having driving point impedance in the Foster-I and Foster-II form. (8)

$$Z(s) = \frac{s(s^2+9)}{(s^2+5)(s^2+13)}$$

- b) Check whether the function (7)

$$F(s) = \frac{s^2+10s+4}{s+2} \text{ is positive real or not.}$$

- Q9 Write short notes on (6)

- a) Positive real Function and its properties (6)
b) Explain Passive filters and their classification (9)
